

Dipayan Chakraborty

Associate Scientist

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RESEARCH INTEREST

Low-energy electron–molecule interactions, Dissociative Electron Attachment (DEA), Velocity Map Imaging (VMI), Electron-induced fragmentation dynamics, Temporary Negative Ions, Electron-driven chemistry, Molecular reaction dynamics, Cold molecular ions, Molecular Cluster Dynamics, Instrumentation development, Mass spectrometry, Gas-phase molecular physics.

PROFESSIONAL EXPERIENCE

Associate Scientist

Feb 2026 – Present

*Heyrovský Institute of the Czech Academy of Sciences
Prague, Czech Republic*

Research Group: Dr. Jaroslav Kočíšek

Research Associate

Jul 2025 – Oct 2025

*Indian Institute of Science Education and Research Kolkata
India*

Research Group: Prof. Dhananjay Nandi

Postdoctoral Research Associate

Jun 2019 – Apr 2025

*Radiation Laboratory, University of Notre Dame
Notre Dame, Indiana, USA*

Supervisor: Prof. Sylwia Ptasinska

Visiting (affiliated) Research Scholar

Sep 2021 – May 2024

*Lawrence Berkeley National Laboratory
Berkeley, California, USA*

Host Scientist: Dr. Daniel S. Slaughter

EDUCATION

Ph.D. in Experimental Atomic and Molecular Physics

2014–2019

Indian Institute of Science Education and Research (IISER) Kolkata, India

Dissertation: *Low-Energy Electron–Molecule Collision Studies in Gas Phase*

Advisor: Prof. Dhananjay Nandi

M.Sc. in Physics

2012–2014

National Institute of Technology Durgapur, India

B.Sc. in Physics

2008–2011

RESEARCH HIGHLIGHTS

- Designed and commissioned an absolute Dissociative Electron Attachment (DEA) spectrometer for quantitative cross-section measurements.
- Designed, constructed, and commissioned a complete Velocity Map Imaging (VMI) spectrometer at the University of Notre Dame.
- Extensive experience in electron–molecule collision experiments, temporary negative ions, and molecular fragmentation dynamics.
- Developed custom LabVIEW-based instrument control and MATLAB data analysis software for charged-particle imaging experiments.
- Lead author of a topical review on ion-pair dissociation dynamics in *J Phys. B: Atom. Mol. Opt. Phys.*
- Reviewer for journals published by AIP and IOP.

TECHNICAL EXPERTISE

Experimental Techniques	Velocity Map Imaging (VMI), Dissociative Electron Attachment (DEA), Ion-Pair Dissociation (IPD), Electron–Molecule Collision Spectroscopy, Quadrupole Mass Spectrometry (QMS), Time-of-Flight Mass Spectrometry (TOF-MS). Complex Cluster Beam Experiments.
Instrumentation	Design and commissioning of charged-particle spectrometers, electron optics, MCP detectors, high- and ultra-high vacuum systems, instrument calibration, vacuum technology.
Data Acquisition	NI LabVIEW, instrument control, data acquisition, electronic interfacing, automation.
Programming	MATLAB, LabVIEW, LaTeX.
Engineering Software	Autodesk Inventor, SolidWorks, SIMION ion trajectory simulations.
Operating Systems	Windows, macOS, Linux (basic).

Publication Summary

Peer-reviewed journal articles	15
First-author publications	8
Corresponding-author publications	4
Review article	1
Conference proceeding	1
Reviewer	Journal of Chemical Physics, Journal of Physics B, Measurement Science and Technology, Journal of Physics D, Physica Scripta

PUBLICATIONS

Lead-Author Publications (Corresponding Author*)

1. **Dipolar Dissociation Dynamics in Electron Collisions with Carbon Monoxide**
Dipayan Chakraborty, Pamir Nag, and Dhananjay Nandi
Phys. Chem. Chem. Phys. **18**, 32973 (2016)
2. **A New Time-of-Flight Mass Spectrometer for Absolute Dissociative Electron Attachment Cross-Section Measurements in Gas Phase**
Dipayan Chakraborty, Pamir Nag, and Dhananjay Nandi
Rev. Sci. Instrum. **89**, 025115 (2018)

3. **Dissociation Dynamics in Low-Energy Electron Attachment to Ammonia Using Velocity Slice Imaging**
Dipayan Chakraborty, Aranya Giri, and Dhananjay Nandi
[Phys. Chem. Chem. Phys.](#) **21**, 21908 (2019)
4. **Absolute Dissociative Electron Attachment Cross-Section Measurements of Difluoromethane**
Dipayan Chakraborty and Dhananjay Nandi
[Phys. Rev. A](#) **102**, 052801 (2020)
5. **Dissociative Electron Attachment to Amide Bond Containing Molecules: N-Ethylformamide and N-Ethylacetamide**
Dipayan Chakraborty, Lauren Eckermann, Ian Carmichael, and Sylwia Ptasinska
[J. Chem. Phys.](#) **153**, 224306 (2020)
6. **Topical Review: Dissociation Dynamics of Ion-Pair States Accessed by Low-Energy Electron Collisions**
Dipayan Chakraborty* and Anirban Paul
[J. Phys. B](#) **56**, 142001 (2023)
7. **Dynamics of Resonant Low-Energy Electron Attachment to Ethanol-Producing Hydroxide Anions**
Dipayan Chakraborty*, Daniel S. Slaughter, and Sylwia Ptasinska
[Phys. Rev. A](#) **108**, 052806 (2023)
8. **Dissociative Electron Attachment to Gas-Phase Acetic Acid Using a Velocity Map Imaging Technique**
Dipayan Chakraborty, Giorgi Kharchilava, Ian Carmichael, and Sylwia Ptasinska
[J. Phys. B](#) **56**, 245202 (2023)
9. **Structural Rearrangements and Fragmentation Pathways Induced by Low-Energy Electron Attachment to Ethyl Acetate**
Anirban Paul, Ian Carmichael, Dhananjay Nandi, Sylwia Ptasinska, and Dipayan Chakraborty*
[Phys. Rev. A](#) **109**, 042818 (2024)
10. **Dissociation of Gas-Phase Anisole Induced by Low-Energy Electron Interactions: Understanding Patterns of Aromatic Bond Cleavage**
Jacob Finley, Giorgi Kharchilava, Ian Carmichael, Dipayan Chakraborty*, and Sylwia Ptasinska
[Phys. Scr.](#) **100**, 045402 (2025)
11. **Fragmentation Dynamics through Geometrical Distortion in Low-Energy Electron Attachment to Carbon Disulfide**
Anirban Paul, Daniel S. Slaughter, Sylwia Ptasinska, Dhananjay Nandi, Ian Carmichael, Jimena D. Gorfinkiel, and Dipayan Chakraborty*
[Phys. Rev. A](#) **111**, 042804 (2025)

Publications from Collaborations

12. **Low-Energy Electron Generation for Biomolecular Damage Inquiry: Instrumentation and Methods**
Elahe Alizadeh, Dipayan Chakraborty, and Sylwia Ptasinska
[Biophysica](#) **2**, 475 (2022)
13. **Low-Energy Electron Driven Reactions in 2-Bromo-5-Nitrothiazole**
Jiakuan Chen, Dipayan Chakraborty, Milan Ončák, Sylwia Ptasinska, and Stephan Denifl
[J. Chem. Phys.](#) **162**, 044304 (2025)

14. **Study of Low-Energy Electron-Induced Dissociation of 1-Propanol**
Soumya Ghosh, **Dipayan Chakraborty**, Anirban Paul, and Dhananjay Nandi
[J. Chem. Phys.](#) **164**, 054302 (2026)
15. **Ion Yields and Velocity Map Imaging of Electron Attachment-Induced Fragmentation of Linear Dialkyl Carbonates: Exploring Dissociation Patterns and Mechanisms**
Giorgi Kharchilava, Jacob Finley, **Dipayan Chakraborty**, Ian Carmichael, and Sylwia Ptasinska
[Eur. Phys. J. D](#) **80**, 71 (2026)

Conference Proceedings

16. **Dipolar Dissociation Dynamics in Electron Collisions with Carbon Monoxide**
Dipayan Chakraborty, Pamir Nag, and Dhananjay Nandi
[J. Phys.: Conf. Ser.](#) **875**, 062008 (2017)

CONFERENCE PRESENTATIONS

Oral Presentations

Resonance Controlled Fragmentation Dynamics in Low-Energy Electron Attachment to Bromo-Triazoles: From Gas-Phase to Microhydrated Environments July 6–9, 2026
15th International Meeting on Atomic and Molecular Physics and Chemistry (IMAMPC-2026), PSI, Switzerland

[Contributed Talk](#)

Development of a Velocity Map Imaging Spectrometer and its Application in Understanding the Dynamics of Low-Energy Electron–Molecule Collisions July 21–26, 2024
27th International Conference on the Application of Accelerators in Research and Industry (CAARI) & the 55th Symposium of Northeastern Accelerator Personnel (SNEAP), Fort Worth, Texas, USA

[Contributed Talk](#)

Dynamics of Resonant Low-Energy Electron Attachment to Dimethylformamide June 3–7, 2024
APS Division of Atomic, Molecular and Optical Physics (DAMOP), Fort Worth, Texas, USA

[Contributed Talk](#)

Abstract ID: C11.00003

Dynamics of Resonant Low-Energy Electron Attachment to Ethanol-Producing Hydroxide Anions August 3–6, 2023
POSMOL 2023, University of Notre Dame, Notre Dame, Indiana, USA

[Invited Talk](#)

Dissociation Dynamics of Low-Energy Electron Attachment to Carbon Disulfide Studied Using Velocity Slice Imaging October 9–13, 2023
APS Gaseous Electronics Conference (GEC), Ann Arbor, Michigan, USA

[Contributed Talk](#)

Abstract ID: FW4.00003

Dissociative Electron Attachment Studies of Molecules Containing Amide Bonds October 5, 2020
APS Gaseous Electronics Conference (Virtual)

[Contributed Talk](#)

Abstract ID: FT1.003

A New Time-of-Flight Mass Spectrometer for Absolute Dissociative Electron Attachment Cross-Section Measurements April 10–13, 2018
3rd International Workshop on Dissociative Electron Attachment, Prague, Czech Republic
Hot Topic Presentation

Poster Presentations

Signature of A_1 Symmetry in the Higher Resonance State in Dissociative Electron Attachment to Ammonia December 3–8, 2018
13th Asian International Seminar on Atomic and Molecular Physics (AISAMP-13), TIFR Mumbai and IIT Mumbai, Mumbai, India

Absolute Dissociative Electron Attachment Cross-Section Measurement Studies for Difluoromethane January 6–8, 2018
Conference on Atomic and Molecular Physics (ISAMP-TC7), IISER Tirupati and IIT Tirupati, Tirupati, India

Electron Impact Ionization and Dissociation of Isolated Molecules and Clusters January 3–6, 2017
21st National Conference on Atomic and Molecular Physics (NCAPM-XXI), Physical Research Laboratory, Ahmedabad, India

Polar Dissociation Dynamics in Electron Collisions with Carbon Monoxide January 9–11, 2016
International Conference on Charged Particle Collisions and Electronic Processes in Atoms, Molecules, and Materials (q-PaCE 2016), Indian School of Mines, Dhanbad, India

Professional Service

Journal Reviewer

- Journal of Chemical Physics (AIP)
- Journal of Physics B (IOP)
- Journal of Physics D (IOP)
- Measurement Science and Technology (IOP)
- Physica Scripta (IOP)

Conference Organization

- Local Organizing Committee, POSMOL 2021, University of Notre Dame
- Local Organizing Committee, POSMOL 2023, University of Notre Dame

Awards and Fellowships

- Department of Energy Postdoctoral Research Fellowship, University of Notre Dame (2019)
- PhD Thesis shortlisted (Top 4) for the Indian Society of Atomic and Molecular Physics (ISAMP) Best Thesis Award (2018–2020)
- IISER Kolkata Doctoral Fellowship (2014–2019)
- Qualified Joint Entrance Screening Test (JEST), Department of Atomic Energy, Government of India
- International Student Travel Grant, IISER Kolkata (2018)

RESEARCH EXPERIENCE

IISER Kolkata, India (2014–2019)

- Conducted doctoral research in Experimental Atomic and Molecular Physics at Indian Institute of Science Education and Research (IISER) Kolkata under the supervision of Prof. Dhananjay Nandi.
- Developed a new spectrometer for absolute Dissociative Electron Attachment (DEA) cross-section measurements of biologically relevant molecules.
- Designed, assembled, and commissioned the complete experimental apparatus, including microchannel plate (MCP) detector, vacuum system, data acquisition electronics, and instrument control system.
- Performed SIMION simulations to optimize the spectrometer geometry and electrode voltages.
- Utilized the instrument to investigate low-energy electron-induced fragmentation dynamics of several molecular systems.
- Performed Velocity Map Imaging (VMI) experiments to investigate Dissociative Electron Attachment (DEA) and Ion-Pair Dissociation (IPD) processes.

University of Notre Dame and Lawrence Berkeley National Laboratory, USA (2019–2025)

- Investigated resonant low-energy electron interactions with amide bond-containing molecules using an electron monochromator coupled to a Quadrupole Mass Spectrometer (QMS).
- Studied electron-induced fragmentation dynamics relevant to radiation damage mechanisms in biologically important molecules.
- Designed, constructed, and commissioned a complete Velocity Map Imaging (VMI) spectrometer at the Radiation Laboratory, University of Notre Dame.
- Developed the complete instrument-control software, data acquisition system, MATLAB analysis codes, spectrometer calibration procedures, and experimental workflow.
- Applied the newly developed instrument to investigate electron attachment dynamics in a variety of molecular systems, resulting in multiple peer-reviewed publications.

Research Expertise:

Low-energy electron–molecule collisions, Dissociative Electron Attachment (DEA), Ion-Pair Dissociation (IPD), Velocity Map Imaging (VMI), electron monochromators, Quadrupole Mass Spectrometry, charged-particle imaging, vacuum instrumentation, instrument development, SIMION simulations, LabVIEW, MATLAB.

Teaching and Mentoring

- Teaching Assistant, Physics I, IISER Kolkata (2015)
- Mentored M.S. research projects of Lauren Eckermann, Giorgi Kharchilava, and Jacob Finley at the University of Notre Dame.

References

Available upon request.